Carry is tradable after costs; the stronger reversal signal is not

Benji Alwis · Disrupt Quant 2026 research note · a low-turnover, sector-neutral strategy, frozen and committed (**d03918c**) before a single public-validation run (**450fb72**)

Period (official engine, from cash, net)	Sessions	Ann. return	Sharpe 1× / 1.5× / 2×	Max DD	Turnover/yr
Development, 2021-01 → 2024-06	910	8.8%	1.66 / 1.61 / 1.57	−3.4%	8.3
Walk-forward, 2023-01 → 2024-06 (construction out of sample; signal discovery in-sample)	390	12.6%	2.44 / 2.39 / 2.35	−2.5%	8.3
Public validation, 2024-07 → 2024-12 (single run; −0.7% in total)	132	−1.3%	−0.22 / −0.28 / −0.35	−5.2%	11.1
Reference: equal-weight long-only, development	910	4.2%	0.42 (1×)	−11.0%	2.4

Table 1. Headline results. Each row is a separate simulation that starts from cash and liquidates at the end; returns are annualised.

1. Hypothesis

Two cross-sectional signals emerged from the research. The first was carry: carry_score showed a persistent relationship with next-session returns relative to other assets in the same sector. In a real market, this type of behaviour might resemble a carry or value premium, but because the data are synthetic, I treat it as an empirical feature of the data-generating process rather than a confirmed economic mechanism. The second was short-term reversal: assets that moved strongly relative to their sector tended to partially reverse that move in the following session, a pattern consistent with short-term liquidity effects. Reversal had the stronger predictive signal, but its rapid decay and high turnover made it unprofitable after transaction costs. Carry was weaker statistically but more persistent and therefore more economical to trade. The key result is that predictive strength alone does not determine whether a signal is practically tradable.

2. Evidence

No leakage. Signals were evaluated against the return actually available to a newly requested position, open(t+1)->close(t+1), and all 15 derived fields were independently reconstructed using information available by time t. **Signal screen.** Of 37 raw and sector-neutral tests across 19 fields, only two survived Bonferroni correction: return_1d (IC −0.054, t=−5.84), implying a reversal signal S=−return_1d with IC +0.054, and carry_score (IC +0.030, t=4.32). return_5d survived Benjamini–Hochberg only and was rejected. **Robustness.** Placebo tests collapsed as expected (lagged-signal IC −0.007; shuffled-return IC −0.001), and removing any single asset changed the reversal IC by at most 6.8%. A dominant common market mode and significant sector covariance motivated dollar- and sector-neutral construction. **Decay.** Reversal loses predictive power after one session, whereas carry remains informative out to 20 sessions. **Caveats.** Both signals were weak in 2021 (reversal IC 0.018, t=1.05), and variance-ratio tests reject nothing, so multi-horizon reversion is not claimed.

3. Construction

Portfolio. At each close, I take the 21-session mean of carry_score, rank assets within each four-name sector, centre the ranks, and scale the portfolio to 120% gross. This makes the book dollar- and sector-neutral at rebalance. A sector is rebalanced only when any held weight has drifted more than a quarter of a rank step from its target, reducing unnecessary turnover. **Parameter choice.** Gross exposure is treated as a risk budget rather than an optimised parameter: development Sharpe is 1.66 / 1.66 / 1.67 at 0.8 / 1.2 / 1.6 gross. The configuration and expected results were timestamped before the 2023–24 tests; later deviations are documented. The 21-session smoothing rule was chosen using 2021–22 only: 58% of daily carry-rank changes reversed within five sessions, while smoothing reduced rank changes from 70% to 24% without materially changing IC. It therefore reduces turnover rather than improving prediction. **Rejected.** A cost-gated reversal overlay added return in 1 of 3 blocks (full-period increment +1.0%/yr, t = 0.64) and became negative at 1.5× and 2× costs, so it was disabled.

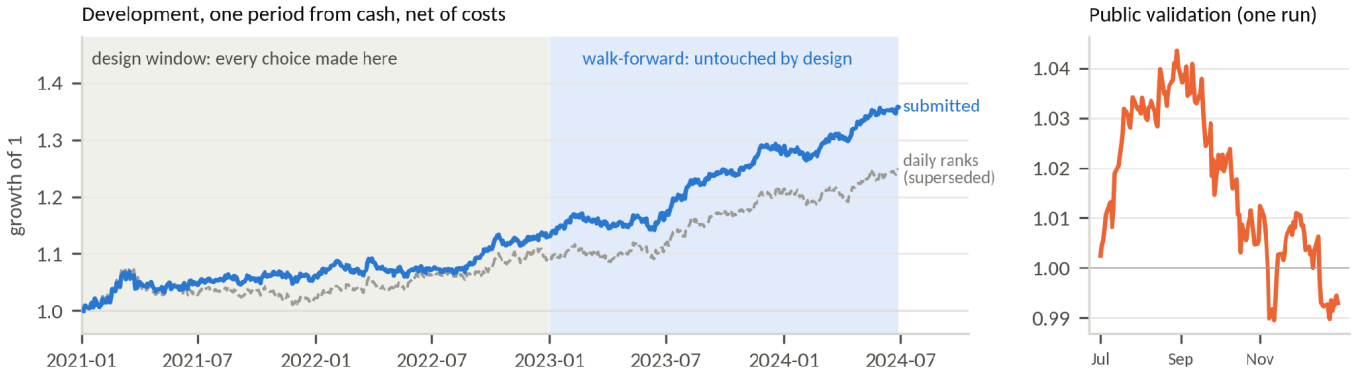


Figure 1. The submitted strategy (blue) and the superseded daily-rank version (grey, dashed) on development; the single validation run (orange) starts again from cash. Periods are separate simulations and are not concatenated.

4. Risk

Exposure. Development beta to the equal-weight market is 0.015. Between rebalances, weight drift produces at most 2.1% net exposure within any sector and 4.0% overall. The largest single position was 8.7% versus the 20% limit, gross exposure is 120% versus the 200% limit, and the engine never rescaled the portfolio. Volatility is about 5%. **Diversification.** All six sectors, 16 of 24 assets, and both the long and short books contributed positively in development; the two largest assets accounted for 31% of positive P&L. **Failure modes.** The strategy is vulnerable to weak carry regimes, to genuine rapid changes in carry that the 21-session average follows with a lag, and to real-world stock-borrow and financing costs that the engine does not model.

5. Costs

Execution costs drive the design. An estimated 81–95% of trading cost comes from half-spread, which is approximately linear in trade size; spreads vary 9.6× across assets, while market impact at the challenge scale is relatively small. Reversal predicts almost twice as strongly as carry but turns over about 40 times as much as the submitted strategy, so its gross edge is eliminated by costs. Smoothing carry leaves its predictive content broadly unchanged while sharply reducing trading. On the design window, gross return was similar for daily and smoothed carry (5.7% vs 5.5%), but cost drag fell from 1.83 to 0.57 percentage points. This is why the submitted strategy's Sharpe declines only from 1.66 to 1.57 when transaction costs are doubled.

Book (development 2021-01 → 2024-06, official engine)	**IC (t)**	**Gross**	**Net 1×**	**Sharpe 1× / 2×**	**Turnover/yr**
Reversal: daily rank of −return_1d	+0.054 (5.84)	11.5%	−8.9%	−2.53 / −7.62	344
Carry: daily within-sector rank	+0.030 (4.32)	8.3%	6.3%	1.22 / 0.86	30.7
Carry: 21-session mean, within-sector rank (submitted)	+0.028 (4.01)	9.4%	**8.8%**	**1.66 / 1.57**	8.3

Table 2. Why the strongest signal was rejected. IC is the sector-neutral daily rank correlation with next-session open-to-close return, with Newey–West t-statistics.

6. Validation

Chronology. The strategy was designed on 2021–22, then its construction was tested walk-forward through June 2024. Those blocks are out of sample for the construction rule, but not for discovery of carry_score, which used the full development sample. The strategy was then frozen, committed and run once on public validation; no rule or parameter changed afterward. **Result.** Validation returned −0.7% in total (−1.3% annualised), with Sharpe −0.22 and maximum drawdown −5.2%, reducing confidence in the development Sharpe. Portfolio behaviour remained normal: turnover was 11×/yr versus 8.3× in development, beta was −0.02, and there were no constraint adjustments. The traded carry IC weakened to +0.013 (t=0.63) rather than reversing sign. The −5.2% drawdown was, however, worse than every six-month development window and is a meaningful warning. With only 132 sessions, Sharpe is estimated imprecisely; 2.2% of overlapping development six-month windows were at least this weak, a descriptive comparison rather than a p-value. No rejected strategy was tested on validation, so it remained a genuine holdout.

7. Limitations

Selection risk remains. Carry was discovered on the same development sample used to evaluate it. Treating all 37 screened signals as independent trials gives a deflated Sharpe ratio of 0.84, while the 95% stationary-bootstrap Sharpe interval is [0.75, 2.60]. The idea of smoothing carry was prompted by 2023–24 results from an AI-assisted candidate strategy, although the specific rule was chosen blind on 2021–22; the AI-assisted candidate that prompted it is disclosed in the README. Its additional 2023–24 gross return has no established mechanism, so the full-development Sharpe of 1.66 is a more reasonable benchmark than the walk-forward 2.44, and validation argues for greater caution. The economic meaning of carry_score is unknown, regime dependence is evident, and 132 validation sessions provide limited power to estimate the true Sharpe.

8. Next step

Regime conditioning. The main next question is whether weak carry periods can be identified in advance using observable state variables such as cross-sectional carry dispersion, sector dispersion or market volatility. If so, gross exposure could be scaled according to predicted signal strength and tested under the same walk-forward discipline. A second, cheaper experiment is the stricter reversal cost gate (κ=1.5), which looked better retrospectively but was deliberately not adopted and should therefore be tested only on genuinely unseen data.